



Práctica cero: Mecánica pulmonar

Departamento de Ingeniería Eléctrica y Electrónica, Ingeniería Biomédica

Tecnológico Nacional de México [TecNM - Tijuana], Blvd. Alberto Limón Padilla s/n, C.P. 22454, Tijuana, B.C., México

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Información general



Nombre del alumno: **Ibarra Monroy Ivana Valeria**

Número de control: 23210709

Correo institucional: l23210709@tectijuana.edu.mx

Asignatura: **Modelado de Sistemas Fisiológicos**

Docente: **Dr. Paul Antonio Valle Trujillo; paul.valle@tectijuana.edu.mx**

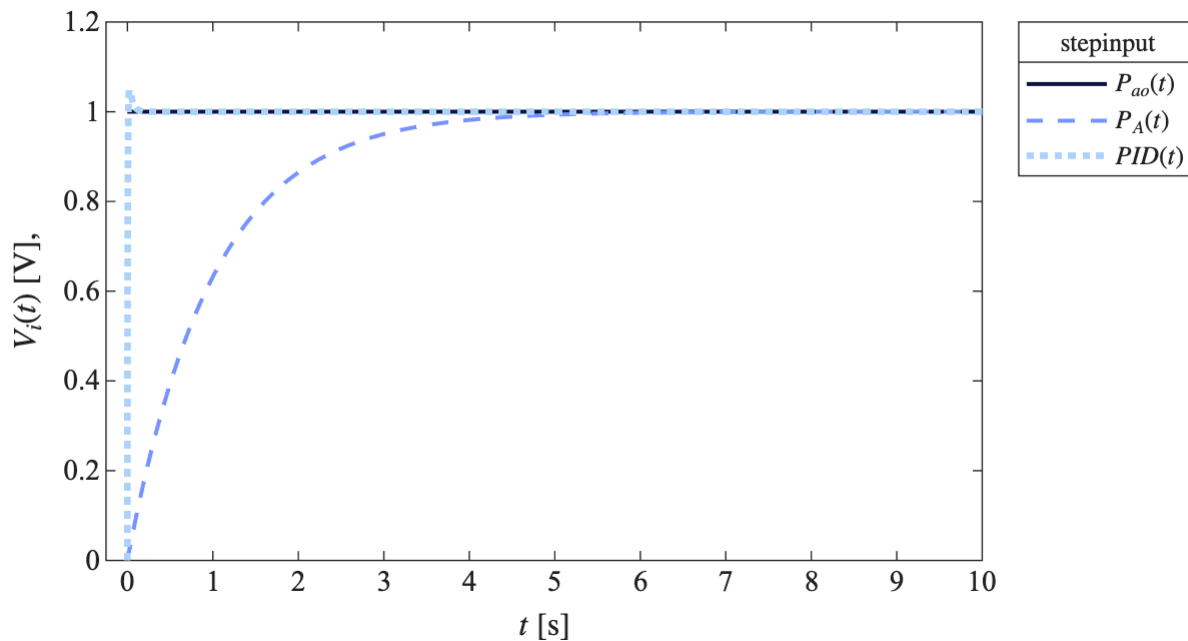
Datos de la simulación

```
clc; clear; close all; warning('off','all')
open_system('ibarra23210709.slx')
parameters.StopTime = '10';
```

```
parameters.Solver = 'ode15s';
parameters.MaxStep = '1e-3';
```

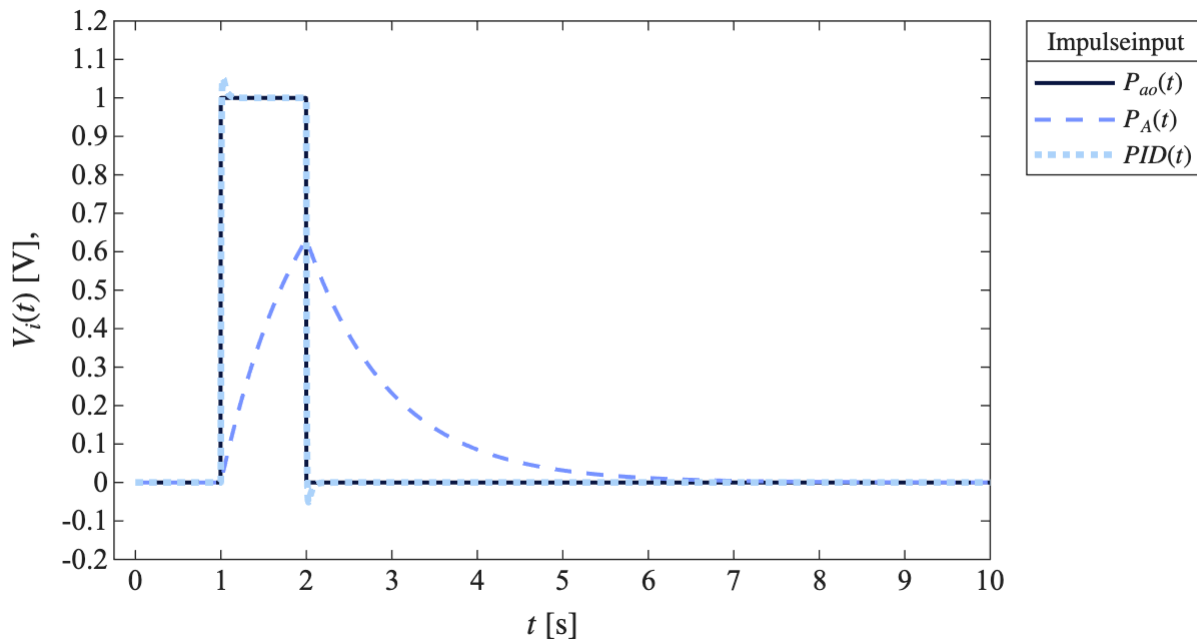
Respuesta al escalón

```
set_param('ibarra23210709/S1', 'sw', '1')
set_param('ibarra23210709/Pao(t)', 'sw', '1')
x1 = sim('ibarra23210709', parameters);
plotsignals(x1.t, x1.Pao, x1.PA, x1.PID, 'step')
```



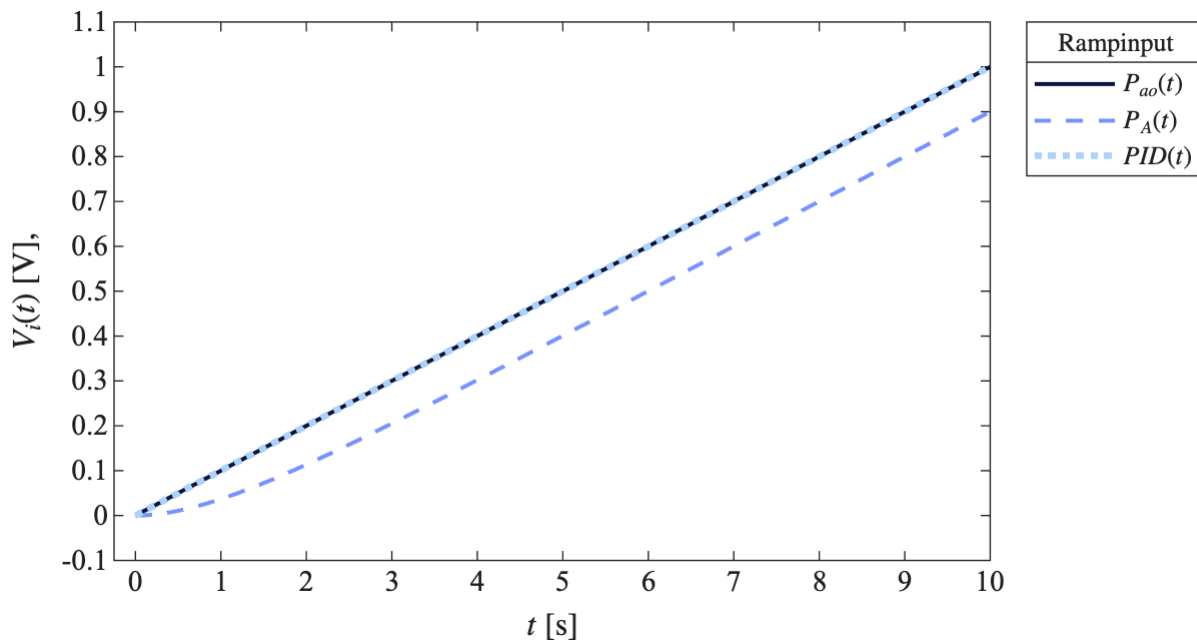
Respuesta al impulso

```
set_param('ibarra23210709/S1', 'sw', '0')
set_param('ibarra23210709/Pao(t)', 'sw', '1')
x2 = sim('ibarra23210709', parameters);
plotsignals(x2.t, x2.Pao, x2.PA, x2.PID, 'Impulse')
```



Respuesta a la rampa

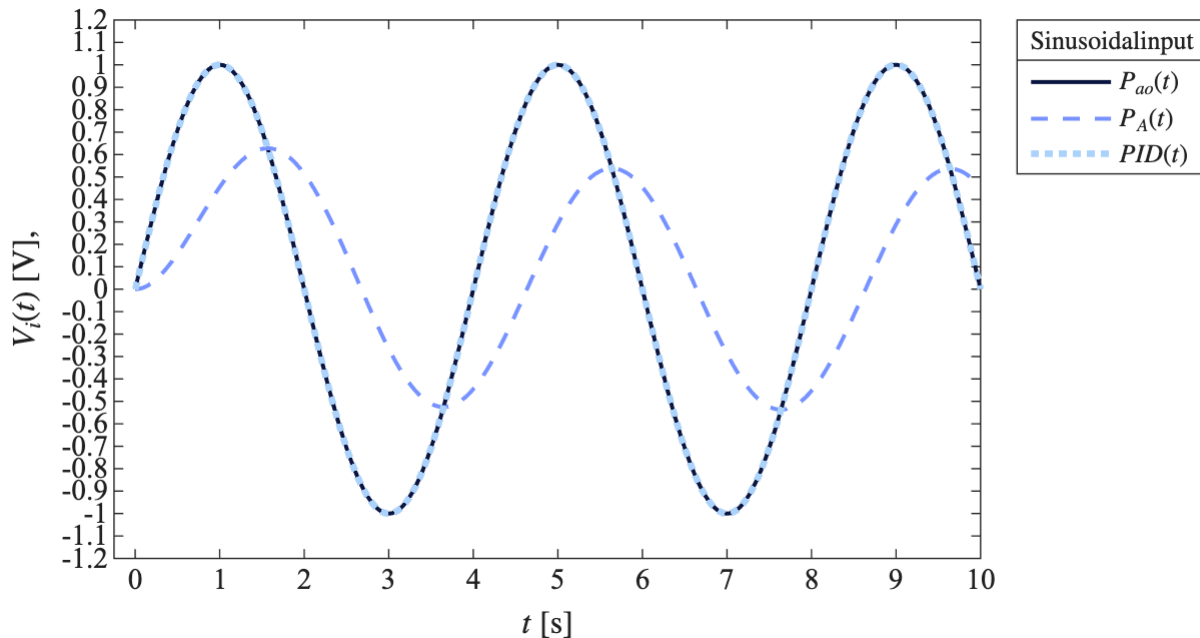
```
set_param('ibarra23210709/S1', 'sw', '1')
set_param('ibarra23210709/Pao(t)', 'sw', '0')
x3 = sim('ibarra23210709', parameters);
plotsignals(x3.t, x3.Pao, x3.PA, x3.PID, 'Ramp')
```



Respuesta a la función sinusoidal

```
set_param('ibarra23210709/S2', 'sw', '0')
set_param('ibarra23210709/Pao(t)', 'sw', '0')
```

```
x4 = sim('ibarra23210709', parameters);
plotsignals(x4.t,x4.Pao,x4.PA,x4.PID, 'Sinusoidal')
```



Funcion: Respuestas de la simulacion

```
function plotsignals(t,Pao,PA,PID,sigal)
set(gcf,'color','w')
set(gcf,'units','inches','position',[1,1,7,3.5])
set(gca,'fontname','times new roman','fontsize',11)
hold on; grid off; box on;
clr1 = [9, 21, 64]/255;
clr2 = [118, 146, 255]/255;
clr3 = [171, 210, 250]/255;
order = [clr1;clr2;clr3]; colororder(order)

plot(t,Pao,'-','LineWidth',1.5,'Color',clr1)
plot(t,PA,'--','LineWidth',1.5,'Color',clr2)
plot(t,PID,':','LineWidth',2.5,'Color',clr3)
L = legend('$P_{ao}(t)$','$P_A(t)$','$PID(t)$');
set(L,'interpreter','latex','location','northeastoutside',...
    'orientation','vertical')
title(L,[sigal,'input'])
xlabel('$t$ [s]','Interpreter','latex')
ylabel('$V_i(t)$ [V]','Interpreter','latex')
xlim([-0.25 10]); xticks(0:1:10)

if sigal == "Step"
    ylim([0 1.2]); yticks (0:0.1:1.2)
elseif sigal == "Impulse"
    ylim([-0.2 1.2]); yticks (-0.2:0.1:1.2)
elseif sigal == "Ramp"
```

```
        ylim([-0.1 1.1]); yticks (-0.1:0.1:1.1)
elseif signal == "Sinusoidal"
    ylim([-1.2 1.2]); yticks (-1.2:0.1:1.2)
end

exportgraphics(gcf,[signal,'_matlab.pdf'],'ContentType','vector')

end
```